

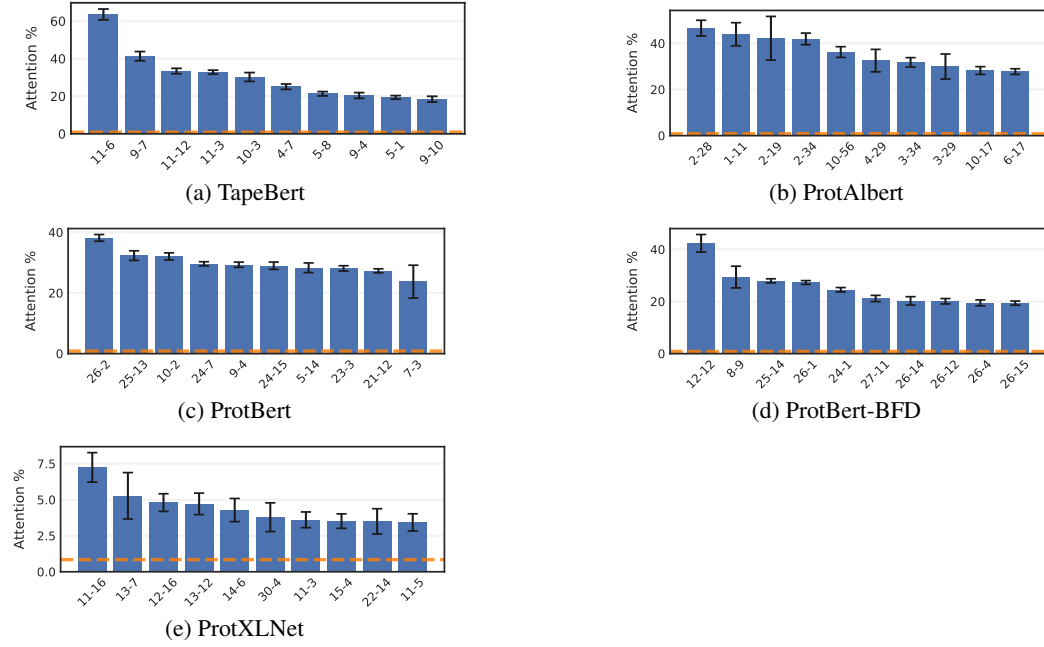


Figure 16: Top 10 heads (denoted by  $\langle \text{layer} \rangle - \langle \text{head} \rangle$ ) for each model based on the proportion of attention focused on PTM positions [95% conf. intervals]. The differences between the attention proportions and the background frequency of PTMs (orange dashed line) are statistically significant ( $p < 0.00001$ ). Bonferroni correction applied for both confidence intervals and tests (see App. B.2).

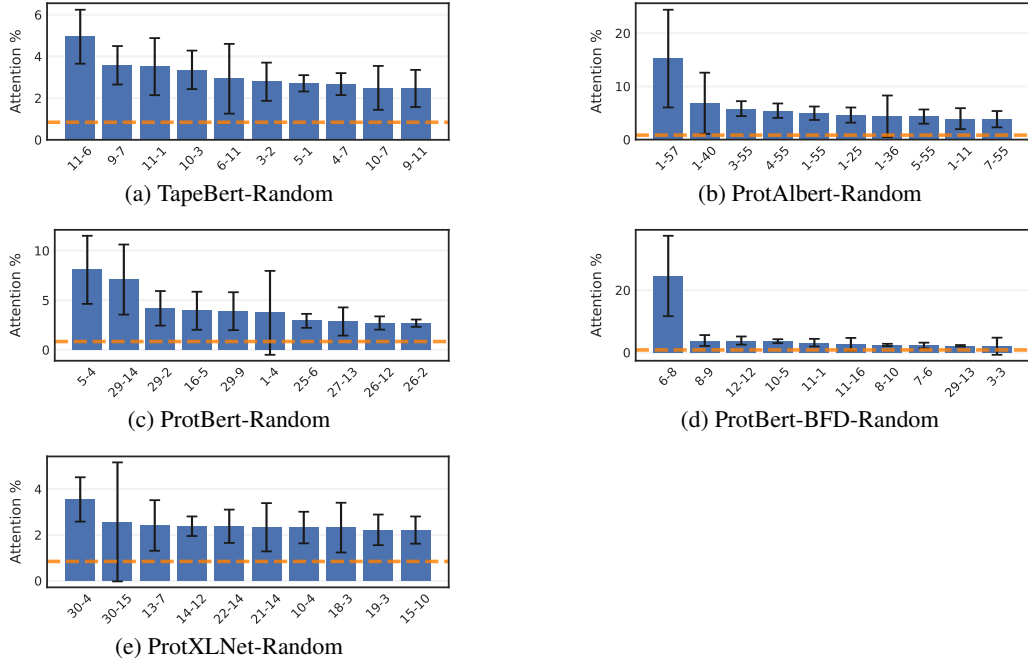


Figure 17: Top-10 heads most focused on PTMs for **null** models. See Appendix B.2 for details.

## C.5 AMINO ACIDS

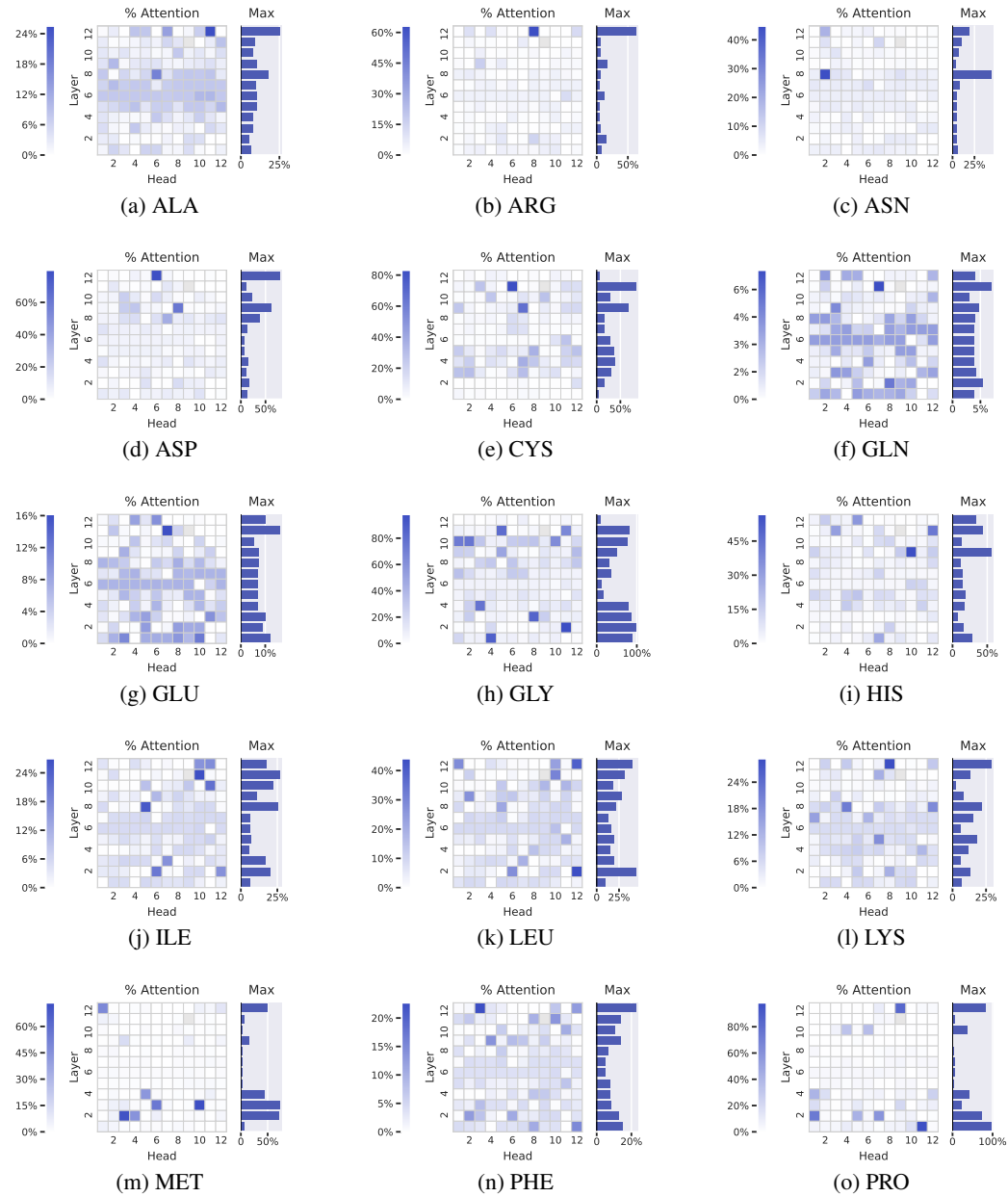


Figure 18: Percentage of each head’s attention that is focused on the given amino acid, averaged over a dataset (TapeBert).

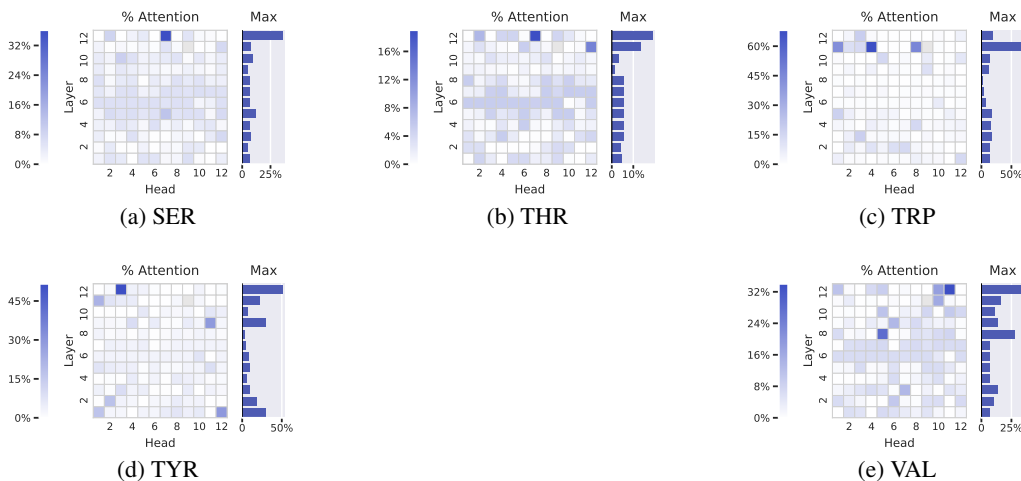


Figure 19: Percentage of each head’s attention that is focused on the given amino acid, averaged over a dataset (cont.)

Table 3: Amino acids and the corresponding maximally attentive heads in the standard and randomized versions of TapeBert. The differences between the attention percentages for TapeBert and the background frequencies of each amino acid are all statistically significant ( $p < 0.00001$ ) taking into account the Bonferroni correction. See Appendix B.2 for details. The bolded numbers represent the higher of the two values between the standard and random models. In all cases except for Glutamine, which was the amino acid with the lowest top attention proportion in the standard model (7.1), the standard TapeBert model has higher values than the randomized version.

| Abbrev | Code | Name          | Background % | TapeBert |             | TapeBert-Random |            |
|--------|------|---------------|--------------|----------|-------------|-----------------|------------|
|        |      |               |              | Top Head | Attn %      | Top Head        | Attn %     |
| Ala    | A    | Alanine       | 7.9          | 12-11    | <b>25.5</b> | 11-12           | 12.1       |
| Arg    | R    | Arginine      | 5.2          | 12-8     | <b>63.2</b> | 12-7            | 8.4        |
| Asn    | N    | Asparagine    | 4.3          | 8-2      | <b>44.8</b> | 8-2             | 6.7        |
| Asp    | D    | Aspartic acid | 5.8          | 12-6     | <b>79.9</b> | 5-4             | 10.7       |
| Cys    | C    | Cysteine      | 1.3          | 11-6     | <b>83.2</b> | 11-6            | 9.3        |
| Gln    | Q    | Glutamine     | 3.8          | 11-7     | 7.1         | 12-1            | <b>9.2</b> |
| Glu    | E    | Glutamic acid | 6.9          | 11-7     | <b>16.2</b> | 11-4            | 11.8       |
| Gly    | G    | Glycine       | 7.1          | 2-11     | <b>98.1</b> | 11-8            | 14.6       |
| His    | H    | Histidine     | 2.7          | 9-10     | <b>56.7</b> | 11-6            | 5.4        |
| Ile    | I    | Isoleucine    | 5.6          | 11-10    | <b>27.0</b> | 9-5             | 10.6       |
| Leu    | L    | Leucine       | 9.4          | 2-12     | <b>44.1</b> | 12-11           | 13.9       |
| Lys    | K    | Lysine        | 6.0          | 12-8     | <b>29.4</b> | 6-11            | 12.9       |
| Met    | M    | Methionine    | 2.3          | 3-10     | <b>73.5</b> | 9-3             | 6.2        |
| Phe    | F    | Phenylalanine | 3.9          | 12-3     | <b>22.7</b> | 12-1            | 6.7        |
| Pro    | P    | Proline       | 4.6          | 1-11     | <b>98.3</b> | 10-6            | 7.6        |
| Ser    | S    | Serine        | 6.4          | 12-7     | <b>36.1</b> | 11-12           | 11.0       |
| Thr    | T    | Threonine     | 5.4          | 12-7     | <b>19.0</b> | 10-4            | 9.0        |
| Trp    | W    | Tryptophan    | 1.3          | 11-4     | <b>68.1</b> | 9-2             | 3.0        |
| Tyr    | Y    | Tyrosine      | 3.4          | 12-3     | <b>51.6</b> | 12-11           | 6.6        |
| Val    | V    | Valine        | 6.8          | 12-11    | <b>34.0</b> | 8-2             | 15.0       |